

Development and Deployment of Software Systems (101)

SPRINT 5 FINAL SUBMISSION

VANTAGE

Technical Manual

Final Version - Deployment Module

Course	Development and Deployment of Software Systems (101)
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Project	VANTAGE - Agile task, sprint, bug, analytics, and Telegram bot platform
Module	Deployment process: install, configure, run, validate, and troubleshoot
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Document Control

Field	Value
Document purpose	Provide the final deployment manual for technical personnel responsible for installing, configuring, running, validating, and maintaining VANTAGE.
Scope	Deployment module only. The manual focuses on OCI/OKE deployment and includes a local execution appendix for development validation.
Primary repository	oci_devops_project, specifically MtdrSpring/backend, MtdrSpring/terraform, MtdrSpring/utils, and Wallet_reacttodogxdmp.
Security posture	Real passwords, Telegram tokens, OCI auth tokens, and wallet files must be handled as secrets. This final manual uses placeholders instead of exposing credentials.

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1. Purpose and Deployment Scope

This technical manual defines the final deployment process for VANTAGE. It is intended for technical personnel who need to prepare the target environment, configure infrastructure and credentials, build the application, deploy it into Oracle Cloud Infrastructure, validate that it is running correctly, and troubleshoot common failures.

The production target described in this manual is Oracle Cloud Infrastructure using Oracle Kubernetes Engine (OKE), Oracle Container Registry (OCIR), Oracle Autonomous Database, Kubernetes secrets, and a public LoadBalancer service. A local execution path is included only for development checks and troubleshooting.

Final version note: This document intentionally avoids hardcoded secrets. Replace placeholder values such as <DB_PASSWORD>, <TELEGRAM_BOT_TOKEN>, and <OCI_USER_OCID> only in the target environment, not inside source code or the submitted manual.

2. System Overview

VANTAGE is a Java Spring Boot and React application for agile project management. It supports users, tasks, sprints, bug reports, productivity analytics, and Telegram bot interaction. The React frontend is built during the Maven package process and copied into the Spring Boot static resources so production runs as one backend container.

Component	Technology	Deployment role
Backend/API	Spring Boot 3.5.6, Java, Maven	Runs REST APIs, business logic, JPA repositories, security configuration, and Telegram bot integration.
Frontend	React	Built with npm during Maven packaging and served by Spring Boot static resources.
Database	Oracle Autonomous Database	Stores VANTAGE users, tasks, task assignees, sprints, bugs, and bot/vector-related data.
Container	Docker image from MtdrSpring/backend/Dockerfile	Packages target/MyToDoList-0.0.1-SNAPSHOT.jar and exposes port 8080.
Orchestration	Oracle Kubernetes Engine	Runs two application replicas and exposes the app through Kubernetes Services.
Registry	OCI Container Registry	Stores the todolistapp-springboot:0.1 image used by OKE.
Infrastructure	Terraform and shell utilities	Creates and configures OCI networking, database, object storage, OKE, namespace, and secrets.

3. Deployment Architecture

The deployment flow starts with OCI infrastructure provisioning, then builds a Spring Boot executable JAR, creates a Docker image, pushes it to OCIR, and deploys it into OKE. Kubernetes injects database and UI

credentials through secrets and mounts the Autonomous Database wallet into the application container.

Step	Action	Main artifact
1	Provision OCI resources with Terraform and setup scripts.	VCN, gateways, Autonomous Database, Object Storage, OCIR, OKE cluster.
2	Prepare environment state and credentials.	state files, Docker registry login, Kubernetes namespace, dbuser and frontendadmin secrets.
3	Build Spring Boot plus React.	target/MyToDoList-0.0.1-SNAPSHOT.jar.
4	Build and push Docker image.	<DOCKER_REGISTRY>/todolistapp-springboot:0.1.
5	Deploy Kubernetes manifest.	todolistapp-springboot-deployment and LoadBalancer service.
6	Validate externally and internally.	Browser access, API responses, pod logs, database connectivity.

Network path: External users access VANTAGE through the Kubernetes LoadBalancer on port 80. The service forwards traffic to container port 8080, where the Spring Boot application listens. Database connectivity uses the Oracle wallet mounted at /mtdrworkshop/creds.

4. Prerequisites

4.1 OCI access prerequisites

- An OCI tenancy with permission to use Cloud Shell and manage the resources required by the deployment.
- A user added to an OCI group with policies for compute, networking, OKE, database, object storage, and container registry.
- The OCI user OCID available from the OCI Console profile page.
- An OCI auth token for Docker login to OCIR. This is not the OCI console password.
- A target OCI region. The repository examples use the Mexico Queretaro region, but the process works with any supported region.

4.2 Software prerequisites

Tool	Required for	Notes
Git	Cloning the repository	Used in Cloud Shell or local development environment.
Java 22 / GraalVM Java 22	Build and runtime compatibility	env.sh checks Java and sets JAVA_HOME when the expected local GraalVM path exists.

Tool	Required for	Notes
Maven / Maven Wrapper	Spring Boot build	The repository includes mvnw and mvnw.cmd under MtdrSpring/backend.
Node.js and npm	React build	frontend-maven-plugin installs Node v23.9.0 and npm 8.1.2 for the build.
Docker	Image build and push	Must be authenticated with OCIR before pushing.
kubectl	OKE deployment and validation	Configured by setup using oci ce cluster create-kubeconfig.
Terraform	OCI infrastructure	Called by the setup utilities from MtdrSpring/terraform.
OCI CLI	OCI resource operations	Used by setup scripts for database, registry, object storage, and cluster access.

5. Repository Elements Used for Deployment

Path	Purpose
MtdrSpring/env.sh	Loads deployment environment variables, Java path checks, state directory, logs, helper functions, and kubectl aliases.
MtdrSpring/setup.sh	Starts the full setup process and delegates to utils/main-setup.sh.
MtdrSpring/utils/main-setup.sh	Creates or validates OCI state, compartment, Docker registry login, database password secret, UI secret, and background provisioning tasks.
MtdrSpring/utils/oke-setup.sh	Creates TLS files, configures kubeconfig, waits for OKE nodes, and creates the mtdrworkshop namespace.
MtdrSpring/utils/db-setup.sh	Downloads the Autonomous Database wallet, creates the wallet Kubernetes secret, and prepares database objects.
MtdrSpring/terraform/*.tf	Defines OCI network, OKE, Autonomous Database, Object Storage, API Gateway, repository, provider, variables, and outputs.
MtdrSpring/backend/build.sh	Builds the Maven package, builds the Docker image, pushes it to OCIR, and removes the local image after a successful push.

Path	Purpose
MtdrSpring/backend/deploy.sh	Replaces placeholders in the Kubernetes YAML template and applies the deployment in namespace mtdrworkshop.
MtdrSpring/backend/src/main/resources/todolistapp-springboot.yaml	Kubernetes Service and Deployment template for VANTAGE.
MtdrSpring/backend/src/main/resources/application.properties	Spring Boot configuration, Oracle UCP settings, logging, and Telegram bot environment variable defaults.
MtdrSpring/backend/src/main/resources/db_setup.sql	Manual SQL script for VANTAGE schema tables and sequences.

6. Environment Configuration

6.1 Initialize the shell environment

Run the following commands from OCI Cloud Shell or a compatible Linux shell after cloning the repository. The setup scripts are written for bash-style shells.

```
cd oci_devops_project/MtdrSpring
chmod +x *.sh backend/*.sh utils/*.sh
source env.sh
```

Optional: add env.sh to the shell startup file if the same Cloud Shell session will be used repeatedly.

```
echo "source $(pwd)/env.sh" >> ~/.bashrc
```

6.2 Required runtime variables and secrets

Name	Source	Used by
DOCKER_REGISTRY	Generated by setup or state_get DOCKER_REGISTRY	build.sh and deploy.sh image path.
TODO_PDB_NAME	state_get MTDR_DB_NAME or manual export	deploy.sh builds db_url as jdbc:oracle:thin:@<TODO_PDB_NAME>_tp?TNS_ADMIN=/mtdrworkshop/creds.
OCI_REGION	OCI Cloud Shell environment or state_get REGION	Kubernetes manifest environment and OCI commands.
UI_USERNAME	Collected by setup.sh or state_get UI_USERNAME	Injected into the application container.
db_user	Kubernetes manifest value	Database schema user. For VANTAGE, set to VANTAGE.
dbpassword	Kubernetes secret dbuser/dbpassword	OracleConfiguration reads it as the database password.

Name	Source	Used by
db_url	Kubernetes manifest	Oracle JDBC URL with TNS_ADMIN path.
driver_class_name	Kubernetes manifest	Oracle JDBC driver class name.
TELEGRAM_BOT_TOKEN	Environment variable or Kubernetes secret update	Telegram long polling bot authentication. Optional if Telegram is not validated.
TELEGRAM_BOT_NAME	Environment variable	Bot name; defaults to cesarmoran_oci_devops_bot if not set.

6.3 Ports and service exposure

Setting	Value
Application port	Spring Boot runs inside the container on port 8080.
Dockerfile exposure	MtdrSpring/backend/Dockerfile exposes 8080.
Kubernetes service	todolistapp-springboot-service exposes port 80 and forwards to targetPort 8080.
Load balancer	Kubernetes Service type LoadBalancer publishes an external IP for browser access.
Frontend API base	React API.js uses /todolist, so frontend and backend are served from the same origin in production.

7. Database and Wallet Configuration

VANTAGE uses Oracle Autonomous Database through the Oracle JDBC thin driver and a wallet. In OCI deployment, the wallet is mounted into the container as a Kubernetes secret named db-wallet-secret. The application expects the wallet at /mtdrworkshop/creds.

7.1 Database schema

The final VANTAGE schema user is VANTAGE. The relational model includes VANTAGE_USER, SPRINTS, TASK, TASK_ASSIGNEES, BUGS, and sequences for users, tasks, sprints, and bugs. The manual SQL file is located at MtdrSpring/backend/src/main/resources/db_setup.sql.

```
-- Run in Oracle Database Actions or SQL*Plus as an admin user.
-- Replace <DB_PASSWORD> with the secure deployment password.
@mtdrSpring/backend/src/main/resources/db_setup.sql
```

Security: If db_setup.sql contains a placeholder or an old classroom password, replace it before execution. Do not commit the real database password to Git and do not include it in the final submission.

7.2 Wallet configuration

For OCI deployment, `utils/db-setup.sh` generates or downloads the wallet, rewrites `sqlnet.ora` to point to `/mntdrworkshop/creds`, and creates the `db-wallet-secret` Kubernetes secret with `cwallet.sso`, `ewallet.p12`, `keystore.jks`, `ojdbc.properties`, `sqlnet.ora`, `tnsnames.ora`, and `truststore.jks`.

```
kubecttl get secret db-wallet-secret -n mtdrworkshop
kubecttl describe secret db-wallet-secret -n mtdrworkshop
```

7.3 Application datasource resolution

OracleConfiguration resolves datasource values in this order: Spring datasource properties, Kubernetes/container environment variables, and then DbSettings values. The deployment should provide `db_url`, `db_user`, `dbpassword`, and `driver_class_name` through the Kubernetes manifest and secrets.

8. Build Process

8.1 Production build through build.sh

The production build script uses Maven to compile the backend, run the React build through the frontend Maven plugin, repackage the executable Spring Boot JAR, build the Docker image, and push it to OCIR.

```
cd oci_devops_project/MtdrSpring/backend
./build.sh
```

Build item	Value
Image name	todolistapp-springboot
Image version	0.1
Final image	<DOCKER_REGISTRY>/todolistapp-springboot:0.1
JAR copied into image	target/MyToDoList-0.0.1-SNAPSHOT.jar as /tmp/MyToDoList.jar
Runtime command	java -jar MyToDoList.jar

8.2 Manual build command

If `build.sh` needs to be reproduced manually, run the following equivalent command sequence after `DOCKER_REGISTRY` is set and Docker is logged into OCIR.

```
cd oci_devops_project/MtdrSpring/backend
./mvnw -DskipTests clean package spring-boot:repackage
docker build -f Dockerfile -t <DOCKER_REGISTRY>/todolistapp-springboot:0.1 .
docker push <DOCKER_REGISTRY>/todolistapp-springboot:0.1
```

8.3 OCI DevOps build

The repository also includes `build_spec.yaml` for OCI DevOps. It installs GraalVM Java 22, builds the Spring Boot/React package, builds a local Docker image named `todolistapp-springboot:local`, and exports it as the `todolist-image` artifact.

9. OCI/OKE Deployment Procedure

9.1 Clone and prepare the repository

```
mkdir reacttodo
```



```
cd reacttodo
git clone https://github.com/AnaJmz345/oci_devops_project.git
cd oci_devops_project/MtdrSpring
source env.sh
```

9.2 Run infrastructure setup

The setup flow provisions or validates OCI resources, creates the mtdrworkshop namespace, configures kubeconfig, creates database and UI secrets, prepares the wallet secret, and logs progress under the configured log directory.

```
source setup.sh
```

During setup, provide the requested values when prompted: OCI user OCID, optional existing compartment OCID, secure database password, UI username, and UI password. If no compartment is provided, the setup creates one using the generated run name.

9.3 Ensure VANTAGE database user and secret

The Kubernetes deployment template uses db_user=VANTAGE. Confirm the dbuser secret contains the password for the VANTAGE database user.

```
kubectl create secret generic dbuser --from-literal=dbpassword='<DB_PASSWORD>'
--dry-run=client -o yaml | kubectl apply -f - -n mtdrworkshop
kubectl get secret dbuser -n mtdrworkshop
```

9.4 Configure Telegram bot variables

The application reads telegram.bot.token from TELEGRAM_BOT_TOKEN and telegram.bot.name from TELEGRAM_BOT_NAME. For production, provide these as environment variables or a Kubernetes secret patch rather than editing application.properties.

```
kubectl set env deployment/todolistapp-springboot-deployment
TELEGRAM_BOT_TOKEN='<TELEGRAM_BOT_TOKEN>' -n mtdrworkshop
kubectl set env deployment/todolistapp-springboot-deployment
TELEGRAM_BOT_NAME='<TELEGRAM_BOT_NAME>' -n mtdrworkshop
```

Timing: If the deployment does not exist yet, apply the Telegram variables after ./deploy.sh creates it, or add them to the Kubernetes manifest before applying it in the target environment.

9.5 Deploy the application

After the image has been built and pushed, deploy the Kubernetes resources from the backend directory. deploy.sh creates a timestamped YAML file, replaces template placeholders, and applies it to namespace mtdrworkshop.

```
cd oci_devops_project/MtdrSpring/backend
./deploy.sh
```

The generated Kubernetes deployment runs two replicas with imagePullPolicy Always, mounts db-wallet-secret at /mtdrworkshop/creds, reads dbpassword from secret dbuser, reads the UI password from secret frontendadmin, and exposes container port 8080.

10. Running and Accessing VANTAGE

10.1 Confirm the public endpoint

```
kubectl get services -n mtdrworkshop
```

```
kubectl get service todolistapp-springboot-service -n mtdrworkshop
```

Copy the EXTERNAL-IP value for todolistapp-springboot-service and open the following URL in a browser:

```
http://<EXTERNAL-IP>
```

10.2 Local execution for development validation

Local execution is not the production deployment path. Use it only to validate frontend/backend behavior before packaging or to troubleshoot application logic. Configure the Oracle wallet path and Spring datasource properties for the local machine before running.

```
cd oci_devops_project/MtdrSpring/backend
mvn spring-boot:run
# In a second terminal, only if running the React dev server separately:
cd oci_devops_project/MtdrSpring/backend/src/main/frontend
npm install
npm start
```

Access mode	URL
Spring Boot local URL	http://localhost:8080
React dev server URL	http://localhost:3000
Production service URL	http://<EXTERNAL-IP>

11. Deployment Validation

11.1 Kubernetes health checks

```
kubectl get nodes
kubectl get namespace mtdrworkshop
kubectl get pods -n mtdrworkshop -l app=todolistapp-springboot
kubectl rollout status deployment/todolistapp-springboot-deployment -n mtdrworkshop
kubectl get deployment todolistapp-springboot-deployment -n mtdrworkshop
```

11.2 Application smoke tests

Run these checks after the LoadBalancer external IP is assigned. Successful responses confirm that the container is reachable and the API is serving requests.

```
curl -I http://<EXTERNAL-IP>
curl http://<EXTERNAL-IP>/todolist
curl http://<EXTERNAL-IP>/tasks
curl http://<EXTERNAL-IP>/sprints
```

11.3 Logs and database validation

```
kubectl logs deployment/todolistapp-springboot-deployment -n mtdrworkshop --tail=100
kubectl describe pod -n mtdrworkshop -l app=todolistapp-springboot
```

Validation area	Expected result
Pods	Two replicas are Running and Ready.
Rollout	rollout status reports successfully rolled out.

Validation area	Expected result
Service	todolistapp-springboot-service has an EXTERNAL-IP and port 80/TCP.
Frontend	The VANTAGE interface loads in the browser without missing static assets.
API	REST endpoints such as /todolist, /tasks, /sprints, /bugs, and /users respond.
Database	Logs do not show missing Oracle JDBC URL, invalid credentials, missing wallet, or ORA connection errors.
Telegram	When TELEGRAM_BOT_TOKEN is configured, the bot responds to basic messages or commands.

12. Maintenance, Updates, Rollback, and Undeploy

12.1 Deploying an update

For a normal application update, rebuild and push the image, redeploy the manifest, and watch the rollout. Because the manifest uses imagePullPolicy Always with tag 0.1, confirm the image in OCIR has been updated before rollout.

```
cd oci_devops_project/MtdrSpring/backend
./build.sh
./deploy.sh
kubectl rollout status deployment/todolistapp-springboot-deployment -n mtdrworkshop
```

12.2 Restarting or scaling

```
kubectl rollout restart deployment/todolistapp-springboot-deployment -n mtdrworkshop
kubectl scale deployment/todolistapp-springboot-deployment --replicas=2 -n mtdrworkshop
```

12.3 Rollback

If a new deployment fails after a previous version was running, roll back the Kubernetes deployment and confirm pod health. For stronger release control, use unique image tags instead of reusing 0.1.

```
kubectl rollout history deployment/todolistapp-springboot-deployment -n mtdrworkshop
kubectl rollout undo deployment/todolistapp-springboot-deployment -n mtdrworkshop
kubectl rollout status deployment/todolistapp-springboot-deployment -n mtdrworkshop
```

12.4 Undeploying the application

To remove the application deployment and its public service, run the repository undeploy script. This stops the application but does not delete the full OCI infrastructure or database.

```
cd oci_devops_project/MtdrSpring/backend
./undeploy.sh
```

13. Troubleshooting Guide

Symptom	Likely cause	Action
build.sh says DOCKER_REGISTRY env variable needs to be set	env.sh/setup.sh did not load state or registry was not created.	Run source env.sh, confirm state_get DOCKER_REGISTRY, or export DOCKER_REGISTRY manually.
Docker push denied or anonymous users read-only	Docker is not authenticated with OCIR.	Create an OCI auth token and run docker login <region>.ocir.io with <namespace>/<username>.
Maven build fails on Java version	JAVA_HOME or PATH points to an incompatible JDK.	Run java -version, source env.sh, and ensure Java 22/GraalVM Java 22 is available.
npm or frontend build failure	Frontend dependencies did not install or Node/npm environment failed.	Re-run ./mvnw -DskipTests clean package and inspect frontend-maven-plugin logs.
ImagePullBackOff	OKE cannot pull from OCIR or image path is wrong.	Check DOCKER_REGISTRY substitution, image exists in OCIR, and cluster has pull permissions.
CrashLoopBackOff	Application starts then fails, often due to database configuration.	Run kubectl logs; verify db_url, db_user, dbpassword, driver_class_name, and db-wallet-secret.
Oracle JDBC URL is missing	OracleConfiguration did not receive datasource or env variables.	Set SPRING_DATASOURCE_URL or db_url and redeploy/restart pods.
ORA-01017 invalid username/password	dbuser secret does not match the VANTAGE database user password.	Update dbuser secret and restart the deployment.
ORA wallet or TNS error	Wallet secret missing, sqlnet.ora points to wrong path, or TNS name does not match database service.	Confirm db-wallet-secret exists, is mounted at /mtdrworkshop/creds, and db_url uses <PDB>_tp.
LoadBalancer external IP pending	OKE load balancer provisioning is still in progress or quota/networking is unavailable.	Wait several minutes, describe the service, and check OCI load balancer quotas and events.
Browser loads but API calls fail	API route/base path mismatch or backend unavailable.	Verify frontend API.js uses /todolist and test http://<EXTERNAL-IP>/todolist directly.
Telegram bot does not respond	Token/name missing, invalid token, or bot container logs show Telegram errors.	Set TELEGRAM_BOT_TOKEN and TELEGRAM_BOT_NAME, restart deployment, and check logs.

14. Final Acceptance Checklist

- OCI group, policies, and user access are confirmed.
- Cloud Shell or deployment shell is using the expected architecture and required tools.
- env.sh has been sourced and setup.sh completed successfully.
- OKE namespace mtdrworkshop exists and kubectl can access the cluster.
- Autonomous Database is active and the VANTAGE schema has been created.
- dbuser, frontendadmin, and db-wallet-secret Kubernetes secrets exist in namespace mtdrworkshop.
- The Spring Boot/React package builds without errors.
- Docker image <DOCKER_REGISTRY>/todolistapp-springboot:0.1 exists in OCIR.
- todolistapp-springboot-deployment has two Running/Ready replicas.
- todolistapp-springboot-service exposes port 80 with an EXTERNAL-IP.
- The VANTAGE web UI loads in a browser from the external IP.
- API endpoints respond and can read/write expected database-backed data.
- Logs show no missing datasource, wallet, image pull, or Telegram token errors.
- Rollback and undeploy commands are documented for maintenance personnel.

Submission status: This is the final version of the VANTAGE technical manual deployment module. It contains the cover page and the complete deployment process required for Sprint 5.